

Code Sentinel

Project Plan

1. Overview and Tools Used

This document is the project plan for Code Sentinel, a multi-agent code review platform. The plan covers 17 weeks, from 27-Jul-2026 to 22-Nov-2026, and is divided into 8 work packages containing 30 tasks and 10 milestones. Every milestone is aligned with a course deliverable deadline. The team has 8 members and the planned effort is 200 person-hours per member, which is 1600 person-hours in total.

Tool	What it is used for	File in repo
GanttProject 3.3	Task timeline, milestones, dependencies and per-member resource allocation.	CodeSentinel_Project_Plan.gan
Trello	Kanban task board - 6 lists, 40 cards, one card per task with owner, labels and due date.	Docs/Trello_Board.png
Git repository	Both files and this document are version controlled and submitted in the repo.	/docs folder

2. Project Timeline and Milestones

The chart below shows the 8 work packages across the 17 weeks. The diamonds mark the milestones M1 to M10. Where two milestones fall within a couple of days of each other the markers are spaced apart slightly so both stay readable; the exact dates are given in the table.

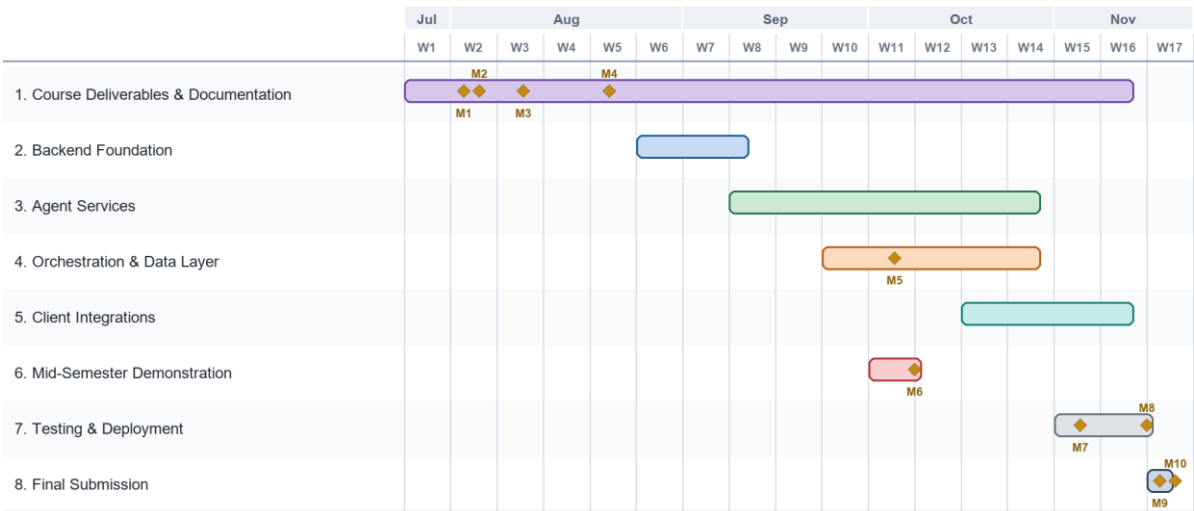


Figure 1: Work package timeline with milestones M1 to M10

ID	Milestone	Planned date	Week	Course deliverable it satisfies
M1	Project Proposal submitted (due 5-Aug)	05-Aug	W2	Deliverable 1 - Project Proposal
M2	Project Plan submitted (due 9-Aug)	07-Aug (due 9-Aug)	W2	Deliverable 2 - Project Plan
M3	Requirements Spec submitted (due 16-Aug)	14-Aug (due 16-Aug)	W3	Deliverable 3 - Requirements Spec (BRD)
M4	System Design Document submitted (27-Aug)	27-Aug	W5	Deliverable 4 - System Design Document
M5	Core pipeline working (gateway + orchestrator + 2 agents)	09-Oct	W11	Internal checkpoint (pre-demo)
M6	Mid-semester demonstration (week of 12-Oct)	12-Oct	W12	Deliverable 6 - Mid-Semester Demonstration

ID	Milestone	Planned date	Week	Course deliverable it satisfies
M7	Test Plan submitted (week of 2-Nov)	06-Nov	W15	Deliverable 7 - Test Plan & Test Cases
M8	Final application deployed, live URL (week of 16-Nov)	16-Nov	W17	Deliverables 8 & 9 - Final Application, Deployment
M9	Final report & presentation (18-Nov)	18-Nov	W17	Deliverables 10 & 11 - Final Report, Presentation
M10	Peer review submitted (19-Nov)	19-Nov	W17	Deliverable 12 - Peer Review

3. Task Timeline - Work Breakdown

The full task list is given below, grouped by work package. Duration is in working days. Tasks marked "Whole team" are the course deliverables, which all 8 members contribute to.

Task	Dates	Weeks	Days	Owner / coordinator
1. Course Deliverables & Documentation				
Project proposal document	27-Jul - 05-Aug	W1-W2	8	Whole team
Project plan - Gantt chart & task board	03-Aug - 07-Aug	W2	5	Whole team
Requirements specification (BRD), user stories, wireframes	10-Aug - 14-Aug	W3	5	Whole team
System design - architecture, ER diagram, OpenAPI, mockups	17-Aug - 27-Aug	W4-W5	9	Whole team
Weekly progress reports	27-Jul - 13-Nov	W1-W16	80	Whole team
2. Backend Foundation				
Monorepo (Turborepo) & CI setup	31-Aug - 02-Sep	W6	3	Vatsal Kamani
API Gateway - Express + TypeScript (auth, routing, webhooks)	03-Sep - 11-Sep	W6-W7	7	Ayush Soni
PostgreSQL schema & migrations	14-Sep - 16-Sep	W8	3	Ayush Soni
3. Agent Services				
Style Agent - language dispatcher, Docker sandbox, ESLint/Ruff	14-Sep - 25-Sep	W8-W9	10	Nevil Nandasana
Security Agent - SAST & secret detection	14-Sep - 25-Sep	W8-W9	10	Tej Prakash Tak
Logic Agent - bugs & edge cases	19-Oct - 23-Oct	W13	5	Tej Prakash Tak
Performance Agent - hot path checks	19-Oct - 23-Oct	W13	5	Parin Makwana
Documentation Agent - docstring suggestions	26-Oct - 30-Oct	W14	5	Parin Makwana
4. Orchestration & Data Layer				
LangGraph workflow - parallel fan-out & timeout handling	28-Sep - 06-Oct	W10-W11	7	Sumit Goyal
Aggregation, de-duplication & severity ranking	28-Sep - 02-Oct	W10	5	Durgesh
LLM abstraction layer & provider fallback chain	07-Oct - 09-Oct	W11	3	Sumit Goyal
Vector DB integration - embeddings & similarity search	19-Oct - 30-Oct	W13-W14	10	Durgesh
5. Client Integrations				
GitHub App - webhooks, Octokit, Checks & Review API	19-Oct - 30-Oct	W13-W14	10	Vatsal Kamani
VS Code extension - commands & review trigger	02-Nov - 06-Nov	W15	5	Nevil Nandasana
VS Code webview panel UI	02-Nov - 06-Nov	W15	5	Moksh Mehta
Next.js dashboard, review history & reports	09-Nov - 13-Nov	W16	5	Moksh Mehta
6. Mid-Semester Demonstration				
Demo preparation - slides, partial demo, code walkthrough	05-Oct - 09-Oct	W11	5	Whole team
7. Testing & Deployment				
Test plan & test cases	02-Nov - 06-Nov	W15	5	Whole team
Unit & integration tests, DB seeding, bug log	02-Nov - 13-Nov	W15-W16	10	Whole team
Deployment - Docker, public host, live URL & CI	09-Nov - 13-Nov	W16	5	Vatsal Kamani

Task	Dates	Weeks	Days	Owner / coordinator
8. Final Submission				
Final application freeze & README	16-Nov - 17-Nov	W17	2	Whole team
Final project report (15-25 pages)	16-Nov - 18-Nov	W17	3	Whole team
Poster, video walkthrough, user manual & API docs	16-Nov - 18-Nov	W17	3	Whole team
Final presentation & demo	18-Nov - 18-Nov	W17	1	Whole team
Peer review & contribution report	19-Nov - 19-Nov	W17	1	Whole team

4. Individual Member Timelines and Task Assignments

Each member owns a specific module. The chart below shows when each member's own module work happens. The shared course deliverables (documentation, testing, mid-semester demo and final submission) involve all 8 members and are shown in Figure 1 instead, so that the individual lanes stay readable.

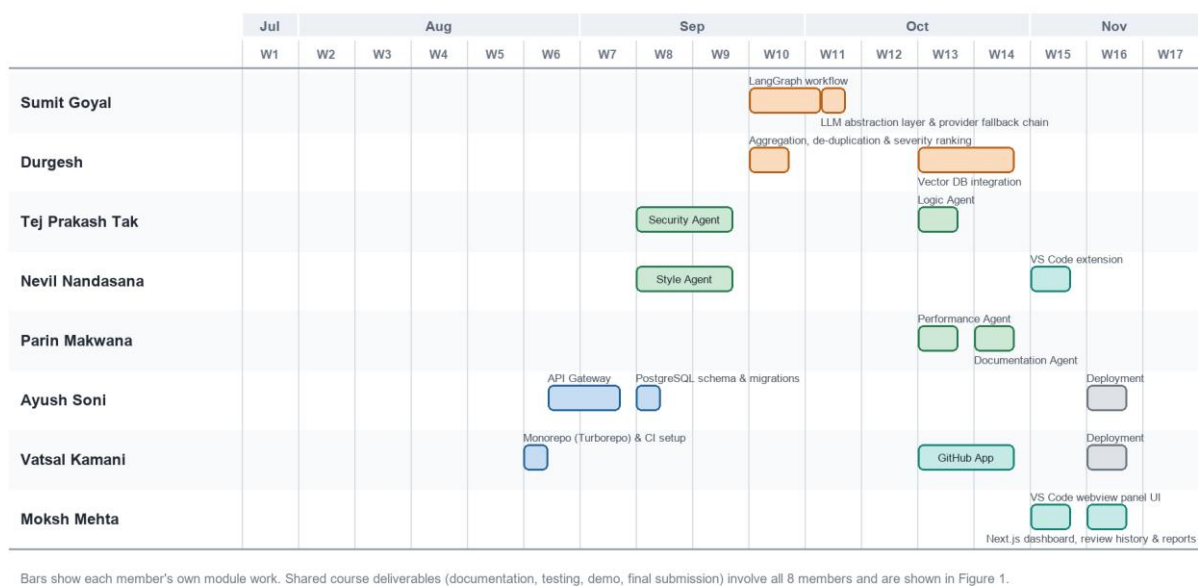


Figure 2: Individual timeline for each team member

Team member	Role	Own module work (with dates)	Hours
Sumit Goyal	Orchestration	LangGraph workflow (28-Sep-06-Oct); LLM abstraction layer & provider fallback chain (07-Oct-09-Oct)	200
Durgesh	Orchestration	Aggregation, de-duplication & severity ranking (28-Sep-02-Oct); Vector DB integration (19-Oct-30-Oct)	200
Tej Prakash Tak	Agent Services	Security Agent (14-Sep-25-Sep); Logic Agent (19-Oct-23-Oct)	200
Nevil Nandasana	Agent Services	Style Agent (14-Sep-25-Sep); VS Code extension (02-Nov-06-Nov)	200
Parin Makwana	Agent Services	Performance Agent (19-Oct-23-Oct); Documentation Agent (26-Oct-30-Oct)	200
Ayush Soni	Backend & Database	API Gateway (03-Sep-11-Sep); PostgreSQL schema & migrations (14-Sep-16-Sep); Deployment (09-Nov-13-Nov)	200
Vatsal Kamani	Integrations & DevOps	Monorepo (Turborepo) & CI setup (31-Aug-02-Sep); GitHub App (19-Oct-30-Oct); Deployment (09-Nov-13-Nov)	200
Moksh Mehta	Frontend	VS Code webview panel UI (02-Nov-06-Nov); Next.js dashboard, review history & reports (09-Nov-13-Nov)	200

In addition to the module work above, every member spends 60 hours on shared work - 20 hours on requirement study and design, 25 hours on integration and testing, and 15 hours on documentation, report and demo. This gives 140 + 60 = 200 hours per member and 1600 hours for the team.

5. Task Board (Trello)

The same 30 tasks and 10 milestones are tracked as 40 cards on a Trello board named Code-Sentinel. Every card is one row of the plan, so the board and the Gantt chart always show the same work. Cards move from left to right as the work progresses, and the two review columns are kept separate so that a task is only counted as done after another member has checked it.

List	Cards	What it holds
Backlog	29	Work not yet started - everything from September onwards.
To Do	7	Next 30 days - requirements spec, system design, monorepo and API gateway.
In Progress	2	Project plan (this document and the Gantt chart), and the weekly progress reports.
In Review	0	Work finished and waiting for review by another team member.
Blocked	0	Work stopped because of a dependency or an unresolved problem.
Done	2	Project proposal document and milestone M1, submitted on 5-Aug.

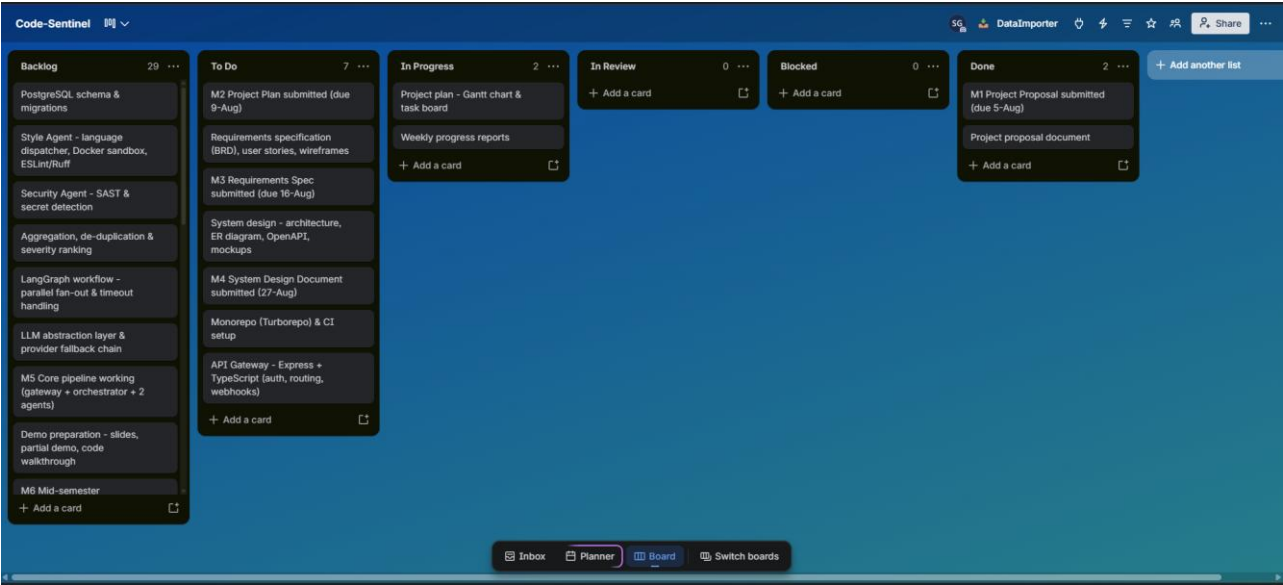


Figure 3: Trello task board as on 8-Aug-2026 (Deliverable 1 complete, project plan in progress)

As on 8-Aug-2026 the board shows 2 cards in Done (the project proposal and milestone M1), 2 cards in progress (this project plan and the weekly progress reports), 7 cards queued in To Do for the next 30 days, and 29 cards still in the backlog. The board is shared with all 8 team members, and each card holds the owner, the planned effort in working days, the tasks it blocks, and a definition of done.